Abstract:

Foundation design in problematic soils presents significant challenges, particularly in mountainous regions characterized by weak strata, high seismic activity, steep slopes, and complex geological conditions. Chenab large-span mountain bridge projects provide valuable insights into innovative foundation solutions developed to overcome these constraints. This study reviews advanced foundation systems such as deep pile foundations, rock-socketed piles, drilled shafts, and anchorage systems adopted to ensure effective load transfer and structural stability. Ground improvement techniques, including grouting, soil reinforcement, and slope stabilization measures, are also discussed to mitigate issues related to settlement, landslides, and differential movement. Advanced site investigation methods combined with numerical modeling techniques enable accurate assessment of soil–rock interaction and foundation performance under static and dynamic loading conditions. The successful implementation of these innovative foundation solutions has significantly enhanced the safety, reliability, and service life of mountain bridge structures constructed in highly challenging terrains. The findings emphasize the importance of integrating geotechnical innovation, advanced construction practices, and site-specific design approaches for sustainable infrastructure development in problematic soils.

Keywords:

Problematic soils; Mountain bridge foundations; Deep pile foundations; Ground improvement techniques.